3.7 Control Architecture

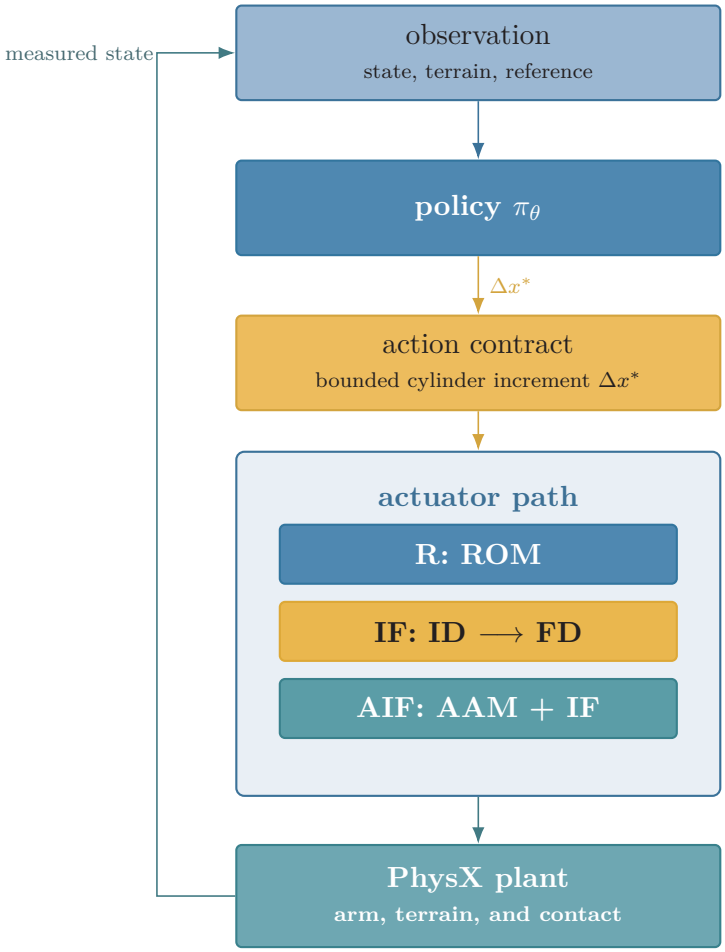


Figure 3.6: Deployed control stack. The policy produces a bounded cylinder increment; R, IF or AIF realizes it in the shared PhysX plant. Measured state closes the loop.

The vehicle advances at the scripted speed and the reference generator supplies the verge centre line and cut height. The policy maps normalized observations to four desired cylinder increments; it has no runtime baseline controller. The action contract bounds those increments. R uses the command ROM, IF the frozen inverse/forward pair, and AIF adds the bounded adapter conditioned on $\hat{\mathbf{z}}$. All paths set the same PhysX cylinder target. The DLS controller is used only for teaching and identification.

3.8 Problem Formulation

The reference is the verge centre line $y_{\mathrm{ref}}(t)$ and cut height h_{ref}, expressed in the truck frame. With cutter-tip position $\boldsymbol{p}_c(t)$ and terrain height $g(\cdot)$, the scored errors are

$$e_{\mathrm{lat}}(t) = p_{c,y}(t) - y_{\mathrm{ref}}(t), \qquad e_{\mathrm{cut}}(t) = \Big(p_{c,z}(t) - g(\boldsymbol{p}_c(t))\Big) - h_{\mathrm{ref}}. \tag{3.7}$$

The objective is to minimize the mean and tail of both errors across path families and terrain while maintaining contact, head alignment, and smooth motion. Learned models are evaluated on one-step and recursive cylinder predictions. The stacks are tested across the randomized plant family; AIF is compared with IF on held-out factors.